Santosh Hanamappa Mokashi

PhD applicant · Computer Architecture & Memory Systems (DRAM)
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RESEARCH PROFILE

Electronics and Communication Engineering graduate (PES University, 2026) working at the intersection of **DRAM/memory systems** and AI hardware. Developed an execution-aware DRAM scheduling architecture for near-memory AI accelerators (Micron Memory Awards, Taiwan) that cut DRAM activity by ~50%, backed by hands-on RTL/SoC design, FPGA prototyping, DFT and SRAM circuit design experience. Seeking a PhD in computer architecture centred on memory-centric computing.

Research interests: DRAM architecture and memory scheduling · near-/in-memory computing for AI · energy-efficient, reliable memory systems · hardware/software co-design

EDUCATION

Dec 2022 – Jul 2026 — **Bachelor of Technology in Electronics and Communication Engineering**
PES University, Bengaluru, India

PUBLICATIONS

Publication — **FPGA-Accelerated RISC-V ISA Extensions for Efficient Neural Network Inference on Edge Devices**

- **System:** custom FPGA-accelerated RISC-V system on Xilinx PYNQ-Z2 with neural-network ISA extensions written in SystemVerilog; timing closure in Vivado.
- **Result:** 2.14× average latency speedup and 49.1% energy reduction, measured on real hardware.

RESEARCH EXPERIENCE

Jan 2026 — **Execution-Aware DRAM Scheduling for Edge-AI Applications** — Micron Memory Awards (Taiwan)
Two-person team project

- **Problem:** the memory wall in AI systems, where DRAM rather than compute limits performance and energy; proposed an execution-aware DRAM scheduling architecture for near-memory AI accelerators.
- **Method:** evaluated with Ramulator (DRAM) and SCALE-Sim (accelerator) simulation under JEDEC-compliant timing constraints; approach compatible with DDR5.
- **Result:** ~50% reduction in DRAM activity and up to 33% energy savings.

Jan 2025 – Jan 2026 — **Project Trainee** — Center for Heterogeneous and Intelligent Processing Systems (CHIPS), PES University
Team project in collaboration with the VLSI Society of India

- **Design:** contributed to the RTL of the AHB protocol micro-architecture for a RISC-V-based SoC for embedded applications.
- **Integration:** integrated an AHB–APB bus architecture connecting three IP blocks, achieving 100 MHz.

PROFESSIONAL EXPERIENCE

Apr 2026 – present — **Intern** — Omni Design Technologies

- **DFT:** implemented the DFT flow for TSMC 28 nm PVT IPs with Cadence Modus, reaching 99.78% stuck-at and 92% at-speed fault coverage.
- **Design & verification:** developing and verifying RTL modules; implementing and validating APB interfaces in Cadence Xcelium with waveform-based debugging in SimVision.

SELECTED PROJECTS

May 2025 — **ASIC design flow for a 64-bit ALU (16 logic functions)** — Cadence

- RTL-to-GDSII 64-bit datapath with a PPA focus: 20% lower power and 10% higher performance.

Apr 2025 — **6T SRAM cell with peripheral circuits** — Cadence Virtuoso

- layout verified with Assura DRC and LVS; read and write operations simulated successfully.

Jan 2024 — **Layered SystemVerilog testbench for a SIPO shift register** — EDA Playground

- directed test sequences and a modular structure (applying Cadence's SystemVerilog for UVM training); 95% code and functional coverage.

HONOURS, FELLOWSHIPS & CERTIFICATES

Sep 2026	**CHIA Hackathon, A³ Workshop @ MICRO 2026** — selected participant; funded GCP/Gemini compute to build agentic loops for HW/SW co-design. Sponsored by Google and NVIDIA.
Mar 2026	**DAC Young Fellow** — 63rd Design Automation Conference (Long Beach, USA); travel grant to present a poster.
Jan 2026	**Micron Memory Awards (Taiwan)** — execution-aware DRAM scheduling project (see Research Experience).
Jan 2025	**Samsung Fellowship** — Cohort 4, India Semiconductor Workforce Development Program (IISc Bangalore, Synopsys, Samsung Semiconductor); selected from 2,500+ applicants.
Certificates	Cadence Design for Test Fundamentals v25.1 · Cadence Essentials of SystemVerilog for UVM (training)

IT & AI SKILLS

Languages	Verilog, SystemVerilog (layered testbenches), C, Python
Memory & AI hardware	Ramulator, SCALE-Sim, DRAM scheduling under JEDEC constraints (DDR5); RISC-V ISA extensions for neural-network inference on edge devices
EDA tools	Cadence Xcelium, SimVision, Genus, Modus, Virtuoso, Assura; Xilinx Vivado
FPGA	Xilinx PYNQ-Z2 prototyping, timing closure in Vivado
Protocols & flows	AMBA AHB, APB, UART; RTL-to-GDSII; DFT

FURTHER EXPERIENCE

Leadership	**Secretary** — IEEE Solid-State Circuits Society (SSCS)

LANGUAGE SKILLS

English	Fluent — language of study and research